

Course Syllabus

18813 / 19746: Foundations of Responsible AI
Fall 2026

Instructor: Sarah H. Cen

Office Location: CIC 2327

Instructor Office Hours: Mondays, 3:30-4:30 in CIC 2327

Location: Posner Hall 153

Number of Units: 12

Pre-requisites/Prior Knowledge: Basic statistics, optimization, and machine learning (familiarity coding in Python needed)

Class days/times: MW 2-3:20 PM

Teaching Assistants: Yifeng Wang, Weichen Yu

TA Office Hours: Wednesdays, 3:30-4:30 in CIC 2206

Teaching staff email: rai-teaching-staff-f26@googlegroups.com

Course Support: Academic Services Center

Office Location: HH 1113

Website: <https://www.ece.cmu.edu/academics/academic-services-center.html>

Course Description: This course provides an interdisciplinary introduction to the key principles and topics in responsible AI, including fairness, interpretability, explainability, privacy, robustness, alignment, and evaluation. Students will develop an understanding of these topics from multiple stakeholder perspectives. The course will cover key methods, drawing on machine learning, statistics, and economics. It will also examine emerging frameworks in AI policy and regulation. Students should have taken an introductory machine learning course or equivalent.

Course Learning Objectives:

By the end of this course, students will be able to:

1. Identify and analyze sources of AI harms and risks;
2. Compare definitions, metrics, and interventions in responsible AI, including their trade-offs and limitations;
3. Understand and apply relevant methods from machine learning, statistics, and economics to address issues in responsible AI;
4. Assess AI systems and interventions from multiple stakeholder perspectives to examine how benefits, harms, and risks are distributed;
5. Analyze and discuss emerging topics and approaches in AI governance;
6. Work collaboratively and communicate research concepts and ideas.

Questions about the course, assignments, project, etc.

Piazza. Please use Piazza for all general questions (e.g., questions about assignments, homework questions, project logistics): <https://piazza.com/class/msddbg3qav61ge> (to see this page, you will need a Piazza account and be enrolled in the course).

Emailing teaching staff. The teaching staff email should *only* be used for personal communications (e.g., a disability accommodation, an academic integrity issue). The teaching staff email is at the top of the syllabus.

Canvas & Gradescope

Canvas. Announcements, course notes, assignments, reading materials, and other course-related documents will be provided on Canvas: <https://cmu.instructure.com/>

Gradescope. Any deliverables (assignments, the project report) will be submitted via Gradescope. A link is provided on the course Canvas page.

- A student guide for Gradescope: <http://www.cmu.edu/teaching/gradescope/>
- Gradescope requires electronic uploads. If you handwrite your submission, you can scan, then upload it (be sure to use a dark pencil or pen!).
- When you upload your work, you will be asked to indicate where each question is located within your submission. Please review these labels before submitting. (Incorrect labels will point graders to the wrong answer, so repeated mislabeling will result in grades of 0 for the corresponding questions.)
- Submissions must also be readable. If the grader cannot read your submission, the default grade will be 0.
- Please keep a soft copy of your assignments for your records.
- If you run into technical issues, email canvas-help@andrew.cmu.edu (although it says canvas-help, they provide tech support for Gradescope as well).

Re-grade policy

You may request re-grades on specific assignment questions. After 2 unsuccessful re-grade requests (one per question), the teaching staff may deny future re-grade requests.

You may request a re-grade of a quiz. However, re-grades will result in re-grading of the entire quiz, and you may receive a lower score than your original score.

Assignments/Assessments

There will be six assignments, two in-person quizzes, and a final project. Please see the course schedule for dates.

- Assignments are due 12 PM (see dates on Schedule)
- For assignments, we may randomly select a subset of questions to grade.
- You are welcome to collaborate with others on assignments, though you must include a mandatory collaboration statement (please see CMU's Academic Integrity Policy and AI Usage Policy below).
- There will be two in-person quizzes. These will be closed-note and fairly short. They are intended to check your understanding at key points in the semester. Assignments during quiz weeks will be shorter, and the assignments will also help you prepare for the quiz.
- The final project will be completed in small groups. There will be a report due before Thanksgiving and short in-class presentations during the last two days of class. There will be small deliverables due in between to help you stay on track (including a short outline and short "brief", which will both serve as the backbone of the final report). In general, grades will be per group, but, if deemed appropriate, students in the same group may receive different grades.

Grading Algorithm:

Assignments: 20%

Quiz #1: 20%

Quiz #2: 20%

Final project: 25%

Participation: 15%

Policy on Late/Make-up Work

Assignments are due 12 PM (see Schedule for dates).

You have 3 late days that can be distributed across the semester. These can be applied to assignments only. For example, you can turn in one assignment 2 days late and another 1 day late. Your late days will be used up in order of the assignments.

Thereafter, 10% (out of the total 100%) will be subtracted per day. This -10% policy applies to the project report as well.

Quizzes and project presentations are in-person. If you anticipate not being able to make it, you must alert the instructor and teaching assistants at least 3 weeks in advance. Make-ups will be provided for emergencies or unavoidable conflicts only. If no make-up arrangement is suitable, you will receive a score of 0 for the quiz or presentation.

Attendance Policy

There is no mandatory attendance. Participation in the course will be considered over the semester and make up 15% of your grade. You are not required to speak publicly in every class, but participation is judged according to contribution to course discussion, breakout discussions, and your group project.

Academic Integrity Policy

You are absolutely permitted to work together on homework assignments and use campus resources (like the Student Academic Success Center, or SASC).

In accordance with the university's policy, you must acknowledge any collaboration or assistance that you receive on work that is to be graded, either from a person, reference, or a tool (including AI tools like ChatGPT). Thus, in any materials you hand in, please include a collaboration/AI usage/resource statement; see [here](#) for examples. (This is a CMU's policy, as per their wording.)

This course will not tolerate plagiarism or cheating. If either is discovered, you will receive a grade of 0 on the corresponding assignment, quiz, or project; no exceptions. Further disciplinary action may be taken. For a clear description of what counts as plagiarism, cheating, and/or the use of unauthorized sources and tools, please see the University's Policy on Academic Integrity (revised in April 2013): <https://www.cmu.edu/policies/student-and-student-life/academic-integrity.html>

You are welcome to approach me in private if you suspect academic integrity issues.

Policy Regarding Use of AI Tools

You may use AI to:

- Understand course materials and assignments
- Brainstorm project proposals and approaches
- Assist in formatting and proofreading assignments or project materials
- Generating code needed for projects that you then verify

You may not use AI as a substitute for understanding, learning, or generating substantive contributions. Regardless of AI usage, your submitted materials will be judged based on their substance and your demonstrated learning. For example, materials generated by AI often “look good” but lack substance or conciseness, and the material may receive a lower grade if the submission fails to show understanding of the material. We reserve the right to ask you to explain any materials that you hand in, such as how you solved a problem or reached a conclusion and to use your response in our evaluations/grading.

Due to university policy (see above), you must include an AI usage statement on any materials handed in. You will not be punished simply for using AI! As stated above, you are being evaluated based on your performance and demonstrated learning.

Recording and Transcription Policy

You may not record or transcribe lectures unless you have a special case permitted by the CMU Office of Disability Resources. Any recording or transcription will be result in immediate loss of the Participation grade (i.e., score of 0 on 15% of the course grade). Recordings permitted by the Office of Disability Resources should not be distributed without my explicit permission or otherwise result in a loss of the Participation grade.

Additional Academic Support Resources

Students are encouraged to avail themselves of the support resources available through the [Student Academic Success Center](#) (SASC). Directions for contacting them are available on their website linked above, or via the contact details below.

Phone: (412) 268-6823

Email: success@andrew.cmu.edu

Location: Posner Hall, 1st floor, open Monday through Friday, 8:30 am to 5:00 pm (EDT)

Accommodations for Students with Disabilities

If you have a disability and require accommodations but do not already have them approved by the Office of Disability Resources, please apply for accommodations through the Application section of the [Disability Resources Online Portal](#). If you already have accommodations approved with Disability Resources, please use the [Accommodations Management](#) section of the Disability Resources Online Portal to notify me about your accommodations, and discuss your accommodations and needs with me as early in the semester as possible. We will work with you to ensure that accommodations are provided as appropriate.

Student wellness

Take care of yourself during the semester. There are many wonderful resources on campus. Do not hesitate at any time to reach out to them. They have helped many students and will likely be able to provide you with assistance as well.

Resources include:

- Counseling and Psychological Services (CaPS): call 412-268-2922 and visit their website at <http://www.cmu.edu/counseling/>
- Re:solve Crisis Network: 888-796-8226
- If the situation is life threatening, call the police
 - On campus: CMU Police: 412-268-2323
 - Off campus: 911

Conduct

All interactions and conversations in this course should be respectful. If you observe any behaviors that you feel are not respectful or believe that I can improve in some way, please feel free to reach out to me privately or report confidentially to the Office for Institutional Equity and Title IX: <https://www.cmu.edu/title-ix/how-to-report-+-options-for-resolution/confidential-reporting.html>

Tentative Course Calendar:

Date	Day	Class Activity	Assignment
August			
24	Mon.	Course introduction, syllabus, discussion	Assignment #1 released
26	Wed.	Topic 1: Fairness - foundations	
31	Mon.	Topic 1: Fairness - methods	
September			
2	Wed.	Topic 1: Fairness - governance	Assignment #1 due Assignment #2 released
7	Mon.	Labor Day – No Class	
9	Wed.	Topic 2: Privacy - foundations	
14	Mon.	Topic 2: Privacy - methods	
16	Wed.	Topic 2: Privacy - governance	Assignment #2 due Assignment #3 released
21	Mon.	Topic 3: Transparency - foundations	
23	Wed.	Topic 3: Transparency - methods	
28	Mon.	Topic 3: Transparency - governance	Assignment #3 due Assignment #4 released
30	Wed.	Quiz #1 (on Topics 1-3)	
October			
5	Mon.	Topic 4: Robustness & Reliability - foundations	
7	Wed.	Topic 4: Robustness & Reliability - methods	
12	Mon.	Fall Break – No Classes	
14	Wed.	Fall Break – No Classes	
19	Mon.	Topic 4: Robustness & Reliability - governance	Assignment #4 due Assignment #5 released
21	Wed.	Topic 5: Alignment - foundations	
26	Mon.	Topic 5: Alignment - methods	
28	Wed.	Topic 5: Alignment - governance	Project roster due Assignment #5 due Assignment #6 released
November			
2	Mon.	No class (meet with your project team)	
4	Wed.	Topic 6: Testing & Evaluation - foundations	
9	Mon.	Topic 6: Testing & Evaluation - methods	
11	Wed.	Topic 6: Testing & Evaluation - governance	Assignment #6, including project proposal due
16	Mon.	Quiz #2 (on Topics 4-6)	
18	Wed.	Special topic (tentatively: Human-in-the-loop)	
23	Mon.	TBD (either special topic or presentation guidelines)	
25	Wed.	Thanksgiving – No Classes	
30	Mon.	Project presentations (Day 1)	Final project due
December			
2	Wed	Project presentations (Day 2)	
7-14		Final Exams Week (no final)	

COURSE OUTCOMES

The ECE Department's BS in ECE is accredited by ABET to ensure the quality of your education. ABET defines 7 Educational Objectives that are fulfilled by the sum total of all the courses you take.

Carnegie Mellon University is accredited by Middlestates to ensure the quality of your education, and asks the department to define Program Learning Outcomes. We have defined 3 major PLOs with sub-bullets for the MS in ECE, plus 1 additional major PLO each for the MS in AIE and for the MS in SE.

Educational Objectives (ABET) or Program Learning Outcomes (Middlestates) represent program-level statements about what students should be able to do upon graduation. The table below maps this course's Course Learning Objectives to the ABET objectives and Middlestates outcomes.

ABET Student Learning Outcomes	Applicable Course Learning Objective
1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	#1, #2, #3
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	#1, #2, #3, #4
3. an ability to communicate effectively with a range of audiences	#3, #4, #5
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	#1, #4
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	#6
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	#3, #4, #6
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	#3, #4, #6
Middlestates Program Learning Outcomes	Applicable Course Learning Objective
I. Demonstrate Technical Expertise in a particular area <i>a. By solving problems in which they apply ECE fundamentals</i> <i>b. By solving complex problems that draw on multiple aspects of ECE</i> <i>c. By solving challenging problems that reflect a depth of understanding in ECE</i>	#3
II. Demonstrate a spirit of Innovation, Collaboration and Leadership	#4, #6

<i>a. By displaying out of the box thinking in solving current complex problems where there are no existing answers</i> <i>b. By successfully collaborating in multidisciplinary teams</i> <i>c. By applying holistic systems-oriented thinking to their designs</i> <i>d. By participating in research projects</i>	
III. Demonstrate professional preparation <i>a. By engaging in job search and internships</i> <i>b. By successfully securing a job as a practicing engineer</i> <i>c. By honing professional skills in teamwork, work organization, and oral and written communication</i>	#6
IV Demonstrate AI Technical and Ethical Expertise (MS-AIE Program) <i>a. By applying artificial intelligence (AI) techniques to solve engineering problems.</i> <i>b. By showing an understanding of the theoretical foundations of machine learning and deep learning techniques.</i> <i>c. By developing and deploying an AI data flow on the cloud using state-of-the-art tools.</i> <i>d. By developing a habit of understanding who may be harmed by AI, and mitigations.</i>	#1, #2, #3, #4, #5